



### 7.1.5 Wrap-Up: Reflection

1. Place an **X** on the line to indicate your mastery of today's lesson.

I can explain why the area of a triangle is half the area of a rectangle with the same base and height for right triangles, acute triangles, and obtuse triangles.

I need help.

I can't get started.

I'm getting there.

I understand and have evidence.



2. Write a brief explanation of your placement of the X.

## Unit 7, Lesson 2: Area of Triangles – Part 2



### Benchmark

MA.6.GR.2.1 Derive a formula for the area of a right triangle using a rectangle. Apply a formula to find the area of a triangle.

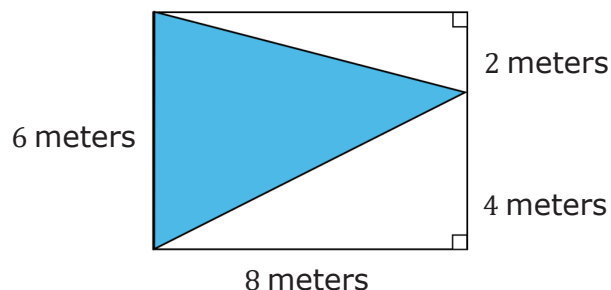


### Learning Target

- ☐ I can apply a formula to find the area of triangles.



### 7.2.1 Warm-Up: Bell Work



1. What is the area of the rectangle? Include units.
2. What is the area of the shaded triangle? Include units.
3. Explain the relationship between the area of a rectangle and triangle that have the same height and base.



### 7.2.2 Guided Instruction: Find the Area of a Triangle

To find the area of any triangle, identify a base of the triangle and its corresponding height.

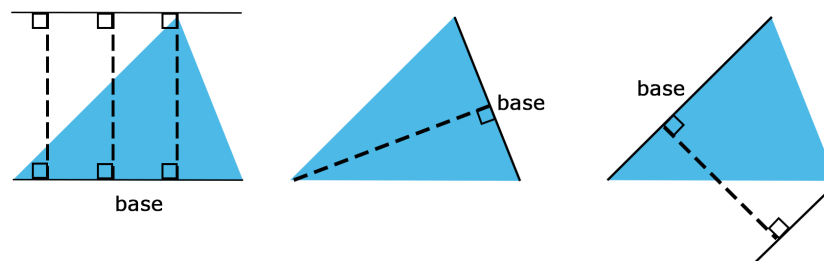


Any of the three sides of a triangle can be the **base**. The term "base" can refer to the side of a triangle and also its measurement.



The corresponding **height** of a triangle is the length of the perpendicular segment drawn from a base to its opposite vertex. The opposite vertex is the vertex of the triangle that is *not* an endpoint of the base.

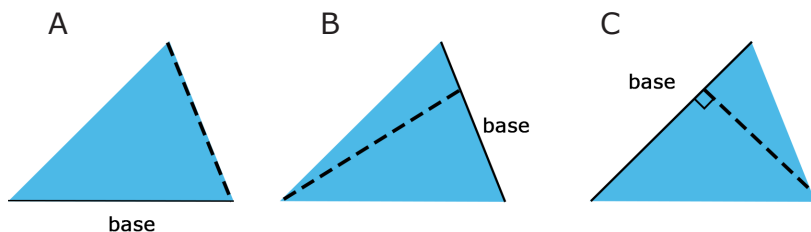
Examples of bases and their corresponding heights on copies of the same triangle are shown. Each dashed line segment represents the height of the triangle.



1. In each example the height is to the base.

- perpendicular
- opposite

2. The side used for the base is labeled on three triangles below. Complete the table to explain why each of the dashed lines do not represent the height of each triangle.

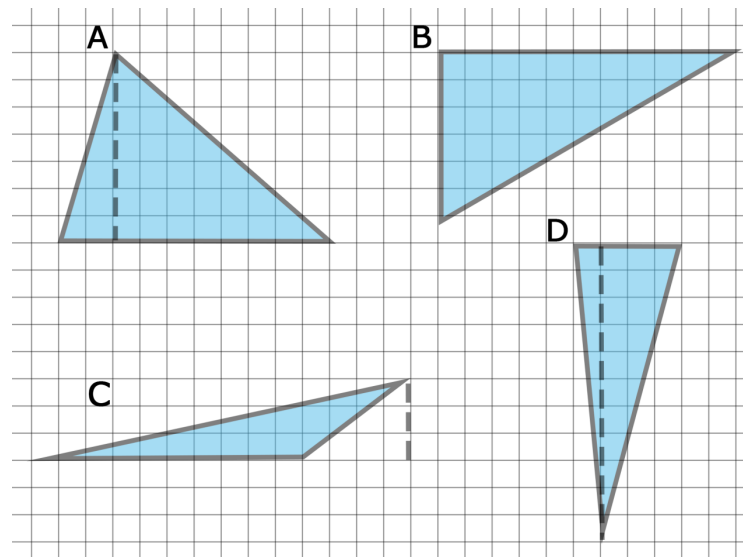


| Triangle   | Explanation |
|------------|-------------|
| Triangle A |             |
| Triangle B |             |
| Triangle C |             |



**Area of a Triangle**  $= \frac{1}{2}bh$ ,  
where  $b$  is the base and  $h$  is the height.

3. Four triangles are shown on the grid. The side length of each square on the grid is 1 centimeter (cm).



- a. For each triangle, label the line segment that is drawn to represent the height,  $h$ . Then, label the side that is the base,  $b$ .

- b. In the table, record the measurement for the base and corresponding height for each triangle. Then, find the area of each triangle. Each square on the grid represents 1 centimeter (cm) by 1 cm.

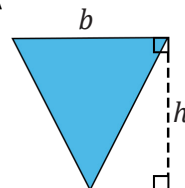
| Triangle     | Base (cm) | Height (cm) | Area (cm <sup>2</sup> ) |
|--------------|-----------|-------------|-------------------------|
| Any Triangle | $b$       | $h$         | $\frac{1}{2}bh$         |
| Triangle A   |           |             |                         |
| Triangle B   |           |             |                         |
| Triangle C   |           |             |                         |
| Triangle D   |           |             |                         |



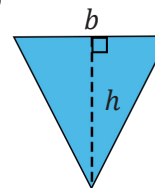
### 7.2.3 Guided Practice: Find the Area of a Triangle

1. Discuss with your partner which drawings correctly label base,  $b$ , and its corresponding height,  $h$ . Circle all the correct drawings.

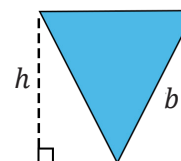
A



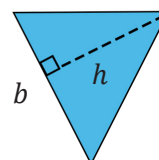
B



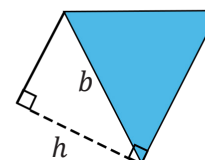
C



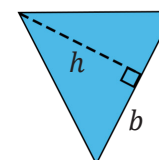
D



E



F



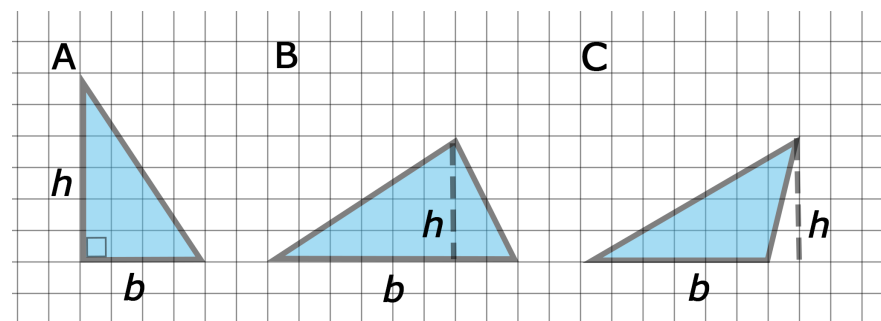
2. For each triangle below, circle the base measurement and put a square around the height measurement. Then, use the base and height to find the area of each triangle. Show your work. Include units.

|  |              |
|--|--------------|
|  | Area = _____ |
|  | Area = _____ |
|  | Area = _____ |
|  | Area = _____ |



### 7.2.4 Your Turn!

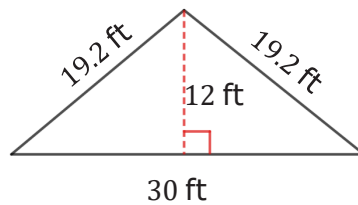
1. For each triangle, a base and its corresponding height are labeled. Complete the table by finding the area of each triangle. The side length of each square on the grid is 1 inch (in.).



| Triangle | Area (in <sup>2</sup> ) |
|----------|-------------------------|
| A        |                         |
| B        |                         |
| C        |                         |

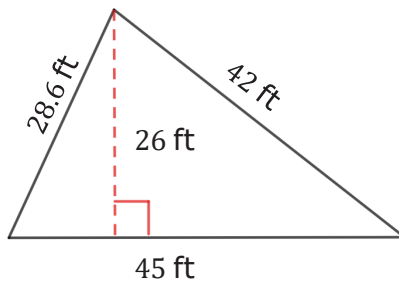
2. Find the area of each triangle. Include units.

a.



Area = \_\_\_\_\_

b.

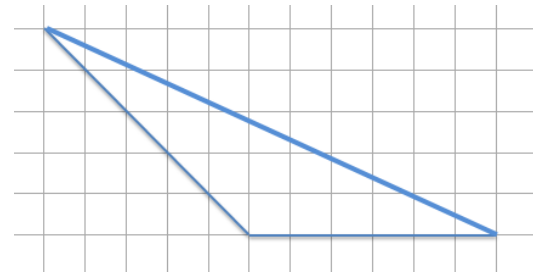


Area = \_\_\_\_\_



### 7.2.5 Wrap-Up: Ticket Out the Door

1. A diagram of a triangle is shown on a grid. The side length of each square on the grid represents 1 foot (ft).



- a. Label a base and its corresponding height for the triangle.
- b. Find the area of the triangle. Include units.